

Problem 287: Adding the Additives

Difficulty: Medium

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Problem Background

Lockheed Martin is always on the lookout for new technology and processes. While 3D printing is relatively new, Lockheed Martin has fully embraced it and is using it to create models, prototypes, and portions of fielded systems, such as airplane fuselages and satellites. When used this way, 3D printing is typically known as Additive Manufacturing. Additive Manufacturing allows Lockheed Martin to manufacture items up to 80% quicker than with conventional techniques while costing substantially less. Unlike home 3D printers, Lockheed Martin's Additive Manufacturing machines use a variety of materials to print with, including metals such as powdered titanium.

The amount of material to be used (and the weight of the object) can be reduced by not printing the object to be completely solid, but by having the inside be printed with a partially hollow pattern. How "solid" the object will be is called the infill percentage; a solid object would have an infill percentage of 100%, but some 3D-printed objects can have an infill as low as 15%.

Problem Description

You have a job as a Lockheed Martin Additive Manufacturing Engineer. You have a lab with printers for different materials and a queue of items to be manufactured. Each item has a design file giving the material to be used, the volume of the item, and the infill percentage. You must determine how many of each item you can print with the material you have.

Since there is virtually no waste in Additive Manufacturing, the mass of material needed to make an item roughly equals the product of the volume of the item, its infill percentage, and the material density. For example, if a material has a density of 3 grams per cubic centimeter (3 g/cm³) and you're printing an object with a volume of 270cm³, and an infill percentage of 70%, you will need 567g of material:

$$270\text{cm}^3 \times 0.7 \times 3\text{g}/\text{cm}^3 = 567\text{g}$$

Sample Input

The first line of your program's input, **received from the standard input channel**, will contain a positive integer representing the number of test cases. Each test case will include:

- A line containing two positive integers separated by a space:
 - N, the number of printers available for use

- P, the number of print jobs queued for processing. This will not exceed N.
- N lines with information about the printers, containing the following values separated by spaces:
 - An uppercase letter, identifying the printer
 - A positive decimal number, representing the density of the material loaded in the printer in g/cm³
 - A positive decimal number, representing the total mass of the material loaded into the printer in grams. This will be greater than the material's density, but not necessarily a multiple of it.
- P lines with information about the print jobs to be processed, containing the following values separated by spaces:
 - An uppercase letter, representing the ID of the printer to handle the job. Each printer will be used at most once within each test case.
 - A positive integer representing the volume of the item in cubic centimeters
 - A decimal between 0 and 1 exclusive representing the infill percentage of the item

```
1
5 3
A 3 2568
B 2.7 2543.4
C 12.45 1540
D 3.2 9000
E 52 403424
A 270 0.7
D 2345 0.8
E 1328 0.25
```

Sample Output

For each test case, your program must print a single line for each print job, containing an integer representing the maximum number of items that can be printed with the provided specifications.

```
4
1
23
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